Mohammed Irtisam Sajin

irtisam.sajin@gmail.com +8801684944952 Portfolio Dhaka, Bangladesh

LinkedIn GitHub Codeforces (Specialist)

EDUCATION

Bangladesh University of Engineering and Technology (BUET) 01/2022 – 06/2026
Bachelor of Science in Electrical & Electronic Engineering Dhaka, Bangladesh
CGPA: 3.65 / 4.00
Focus: Communication & Signal Processing

Notre Dame College Dhaka 06/2018 – 06/2020
Higher Secondary Certificate (HSC), Science, English Version Dhaka, Bangladesh
GPA: 5.00 / 5.00

Rajuk Uttara Model College 02/2016 – 05/2018
Secondary School Certificate (SSC), Science, English Version
GPA: 5.00 / 5.00

PROJECTS

Pathok: A Bangla Handwritten and Scene Text Recognition Deep Learning Model using Grapheme Tokenization
Undergraduate Thesis
Supervisor: Dr. Maruf Ahmed, Assistant Professor, EEE, BUET.
- Developed an OCR model utilizing **CNN** and **Transformer** architectures.
- Integration with custom Bangla character and grapheme-based tokenizers to detect Handwritten and Scene texts from images.
- **Skills & Tools:** Deep Learning, Image Processing, PyTorch, Pandas, NumPy, Kaggle.

Zulip - Full Stack Open Source Contribution
- Reported and fixed an issue on the main Zulip server that was causing errors when forwarding missed messages to user's emails | Issue | PR | **Stack**: Django (Python), Postgresql

Multitask Learning for Segmentation and Classification of Breast Tumors from Ultrasound Images
- Implemented a unified tumor segmentation and classification **deep learning** model for ultrasound breast tumor images.
- Our model achieved **8% higher accuracy** and a **28% higher Dice score** than the standard U-Net model.
- **Tools:** Keras TensorFlow, Scikit-learn, Pandas, NumPy, Kaggle Platform.

Currency Converter Chrome Extension
- A real-time currency converter chrome extension with **200+ active users.** Supports 170+ currencies for conversion. Rates are fetched from **Openexchangerates.org API.**
- **Tools:** JavaScript, HTML, CSS.

Thoughts - A quick and simple tool to save your thoughts
- Used full-stack technologies (MERN) to build an end-to-end **Web application** to quickly save your thoughts.
- Deployed on Vercel (frontend) and Render (backend) | **Live Project** link
- **Tools & Frameworks:** React, Nodejs, Express, MongoDB, Mongoose

Electronic Voting Machine with Fingerprint Authentication
- Developed a master-slave architecture for fingerprint-authenticated vote casting and live vote count tracking. Used **UART** protocol for inter-microcontroller communication.
- **Tools**: Arduino, C Programming, fingerprint sensors, TFT displays.

Mechanical Braille Cell using FPGA Board
- Designed a low-cost refreshable 6-pin Braille cell using servo motors and logic circuits.
- Prototype tested using FPGA board, programmed in **Verilog**, to control the motors and integrated circuits.

Analog Signal Encoding using Delta Modulation and Demodulation
- Simulation and hardware implementation of Delta Modulation and Demodulation, designed using integrated circuits.
- **Hardware:** Breadboard, 555 Timer, D Flip-Flop, and Operational Amplifiers. **Simulation tool:** PSpice.

Three-Body Dynamics: Modeling the Interactions of Point Particles
- 2-D & 3-D motion and energy simulations of three gravitational objects, based on user-defined initial 3D positions and velocities.
- **Tools:** MATLAB and MATLAB GUI.

SKILLS

- **Languages:** C/C++, Python, JavaScript, MATLAB, Assembly and Verilog.
- **Software Frameworks:** TensorFlow, Keras, PyTorch, Scikit-learn, React, NodeJS, Django, Tailwind.
- **Data Analysis:** Pandas, NumPy, SQL, MongoDB.
- **Tools:** PSpice, Arduino, Kaggle, AutoCAD, Quartus, Simulink, Git & Github.
- **Hardware:** Microcontrollers, FPGA, Embedded Systems Programming, Sensor Interfacing.
- Fundamental knowledge of Signal Processing, Machine Learning and Deep Learning.

CERTIFICATES

Machine Learning Specialization - DeepLearning.ai
Got introduced to the fundamentals of Machine Learning and gained practical skills to apply ML in challenging real-world problems.

CS50's Introduction to Programming with Python
Issued by Harvard University (in 2023). Gained practical introduction to Python with a series of classes, completing problem sets and a final project.

REFERENCES

Dr. Shaikh Anowarul Fattah
Professor, Department of EEE, BUET
Email: fattah@eee.buet.ac.bd
Phone: +8801937800854
Official Website

Dr. Maruf Ahmed
Assistant Professor, Department of EEE, BUET
Email: maruf_ahmed@eee.buet.ac.bd
Phone: +8801724874323
Official Website
Thesis Supervisor